

is read for that pixel. Depending on the physical structure of the sensor a colour filter array may be used, which requires a demosaicing/interpolation algorithm. The number of resulting pixels in the image determines its “pixel count”. For example, a 640x480 image would have 307,200 pixels or approximately 307 kilopixels while a 3872 x 2592 image would have 10,036,224 pixels, or approximately 10 megapixels.

The pixel count alone is commonly presumed to indicate the resolution of a camera, but this is a misconception. There are several other factors that impact a sensor’s resolution. Some of these factors include sensor size, lens quality, and the organisation of the pixels.

Many digital cameras have preset modes for different applications. Within the constraints of correct exposure, various parameters can be changed, including exposure, aperture, focussing, light metering, white balance, and equivalent sensitivity. For example a portrait might use a wider aperture to render the background out of focus and would seek out and focus on a human face rather than other image contents.

Common formats for digital camera images are the Joint Photography Experts Group standard (JPEG) and Tagged Image File Format (TIFF).

1.2.2. Scanners

In computing, a scanner is a device that analyzes images, printed text, or handwriting, or an object (such as an ornament) and converts it to a digital image. Most scanners today are variations of the desktop (or flatbed) scanner. The flatbed scanner is the most common in offices. Hand-held scanners, where the device is moved by hand, were briefly popular but are now not used due to the difficulty of obtaining a high-quality image. Both these types of scanners use charge-coupled device (CCD) or Contact Image Sensor (CIS) as the image sensor, whereas older drum scanners use a photomultiplier tube as the image sensor.